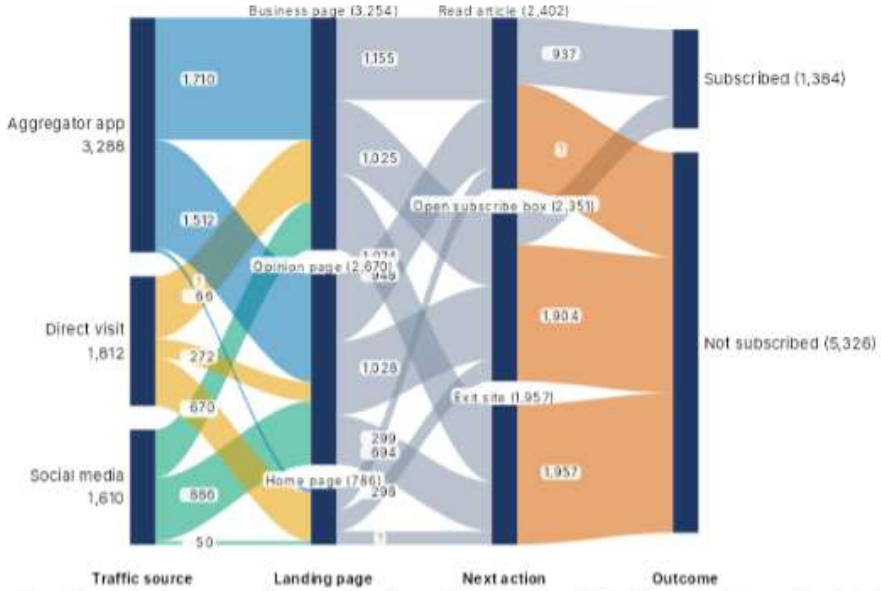
DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot C

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Scenario: The analytics team of a news portal tracked 6,710 visits last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of visits on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Read article to Not subscribed; Home page to Exit site; Direct visit to Business page. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Subscribed as a percentage of all visits; (ii) the largest single flow into "Not subscribed" or "Exit site"; (iii) the percentage of "Opinion page" visits that go on to "Read article". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 300 visits. Compute the height in cm of each landing-page node (Opinion page, Business page, Home page). [2]

(e) Write R code: a links data frame for the first four flows leaving "Direct visit" and "Social media" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

LAB ACTIVITY RECORD

15-10-2026---Activity 6: Website Flow Sankey Design

Department	AIML
Course Code / Course Name	DSA0601 / Data Handling and Visualization
Experiment	Activity 6 - Website Flow Sankey Design
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Aim

To find the three hidden flows of a Sankey diagram of 6,710 website visits by using conservation of flow, to calculate the conversion rate and other percentages from the diagram, to design the final Sankey with reasons for every decision, to work out the node heights for a printed version, to write the R code (networkD3) that draws the whole diagram, and to state when a Sankey diagram misleads.

Key Idea: Conservation of Flow

A Sankey diagram shows how a total number of visits is split and joined again as it moves from one stage to the next. The width of every band is its number of visits. Because no visit appears or disappears between the stages, every node follows one simple rule:

total entering a node = total leaving the node = the number printed on the node

So if all the bands of a node but one are known, the missing one is the node total minus the known bands. This is the rule used in part (a).

Numbers Read from the Diagram

Stage	Node (total visits)
Traffic source	Aggregator app 3,288; Direct visit 1,812; Social media 1,610
Landing page	Business page 3,254; Opinion page 2,670; Home page 786
Next action	Read article 2,402; Open subscribe box 2,351; Exit site 1,957
Outcome	Subscribed 1,384; Not subscribed 5,326

Every stage adds up to the same total: 3,288 + 1,812 + 1,610 = 6,710; 3,254 + 2,670 + 786 = 6,710; 2,402 + 2,351 + 1,957 = 6,710; 1,384 + 5,326 = 6,710. This shows that the node totals are consistent, so the rule above can be trusted. The labelled bands used in the calculations are 1,710, 1,512, 66 (from Aggregator app), 272 and 670 (from Direct visit), 886 and 50 (from Social media), 1,155, 1,025, 1,074 (from Business page), 948, 1,028, 694 (from Opinion page), 299, 298 (from Home page), 937, 1,904 and 1,957.

(a) Hidden Flows by Conservation of Flow [4 marks]

Aim: To find the three flows marked "?" using the rule that what enters a node equals what leaves it.

Working

Hidden flow 1: Read article to Not subscribed

Step 1: Rule used: for the node Read article, visits leaving = visits entering = 2,402.

Step 2: Visits entering Read article: 1,155 (Business page) + 948 (Opinion page) + 299 (Home page) = 2,402. This matches the node total.

Step 3: Visits leaving Read article: 937 go to Subscribed and the rest, x, go to Not subscribed. So 937 + x = 2,402.

Step 4: x = 2,402 - 937 = 1,465.

Step 5: Check: the bands entering Not subscribed are 1,465 + 1,904 + 1,957 = 5,326, which is the node total.

Hidden flow 2: Home page to Exit site

Step 1: Rule used: for the node Home page, visits leaving = 786.

Step 2: Visits leaving Home page: 299 go to Read article, 298 go to Open subscribe box and the rest, y, go to Exit site. So 299 + 298 + y = 786.

Step 3: y = 786 - 299 - 298 = 189.

Step 4: Check: the bands entering Exit site are 1,074 (Business page) + 694 (Opinion page) + 189 (Home page) = 1,957, which is the node total.

Hidden flow 3: Direct visit to Business page

Step 1: Rule used: for the node Direct visit, visits leaving = 1,812.

Step 2: Visits leaving Direct visit: 272 go to Opinion page, 670 go to Home page and the rest, z, go to Business page. So z + 272 + 670 = 1,812.

Step 3: z = 1,812 - 272 - 670 = 870.

Step 4: Check: Business page receives 1,710 (Aggregator app) + 870 (Direct visit) + 674 (Social media) = 3,254. Here 674 = 1,610 - 886 - 50 is what is left of Social media's visits after the Opinion page and Home page bands. The total equals the node total.

Result

Hidden flow	Rule used	Calculation	Value
Read article to Not subscribed	Node total minus the known outflow	2,402 - 937	1,465
Home page to Exit site	Node total minus the known outflows	786 - 299 - 298	189
Direct visit to Business page	Node total minus the known outflows	1,812 - 272 - 670	870

Also found by the same rule: Open subscribe box to Subscribed = 2,351 - 1,904 = 447. The Subscribed node then adds up as 937 + 447 = 1,384, and nothing from Exit site reaches Subscribed.

Complete List of the 23 Flows

From	To	Visits	From	To	Visits
Aggregator app	Business page	1,710	Opinion page	Read article	948
Aggregator app	Opinion page	1,512	Opinion page	Open subscribe box	1,028
Aggregator app	Home page	66	Opinion page	Exit site	694
Direct visit	Business page	870 *	Home page	Read article	299
Direct visit	Opinion page	272	Home page	Open subscribe box	298
Direct visit	Home page	670	Home page	Exit site	189 *
Social media	Business page	674	Read article	Subscribed	937
Social media	Opinion page	886	Read article	Not subscribed	1,465 *
Social media	Home page	50	Open subscribe box	Subscribed	447
Business page	Read article	1,155	Open subscribe box	Not subscribed	1,904
Business page	Open subscribe box	1,025	Exit site	Not subscribed	1,957
Business page	Exit site	1,074			

(b) Calculations [4 marks]

Aim: To calculate the overall conversion rate, the largest single flow into "Not subscribed" or "Exit site", and the share of Opinion page visits that go on to "Read article".

Working and Result

(i) Overall conversion rate

Step 1: Conversion rate = Subscribed visits / all visits x 100.

Step 2: Subscribed = 1,384 and all visits = 6,710.

Step 3: 1,384 / 6,710 = 0.2063, so the rate is 0.2063 x 100 = 20.63%.

Result: The overall conversion rate is about 20.6%. About 1 visit in 5 ends in a subscription.

(ii) Largest single flow into "Not subscribed" or "Exit site"

Step 1: The flows into Not subscribed are 1,465 (Read article), 1,904 (Open subscribe box) and 1,957 (Exit site).

Step 2: The flows into Exit site are 1,074 (Business page), 694 (Opinion page) and 189 (Home page).

Step 3: The largest of all six numbers is 1,957, the flow from Exit site to Not subscribed. As a share of all visits: 1,957 / 6,710 = 0.2917 = 29.17%.

Result: The largest single flow is Exit site to Not subscribed with 1,957 visits (29.2% of all visits). The largest flow into Exit site itself is Business page to Exit site with 1,074 visits (16.0%).

(iii) Percentage of Opinion page visits that go on to "Read article"

Step 1: Opinion page to Read article = 948 visits, and Opinion page total = 2,670 visits.

Step 2: 948 / 2,670 = 0.3551, so the share is 0.3551 x 100 = 35.51%.

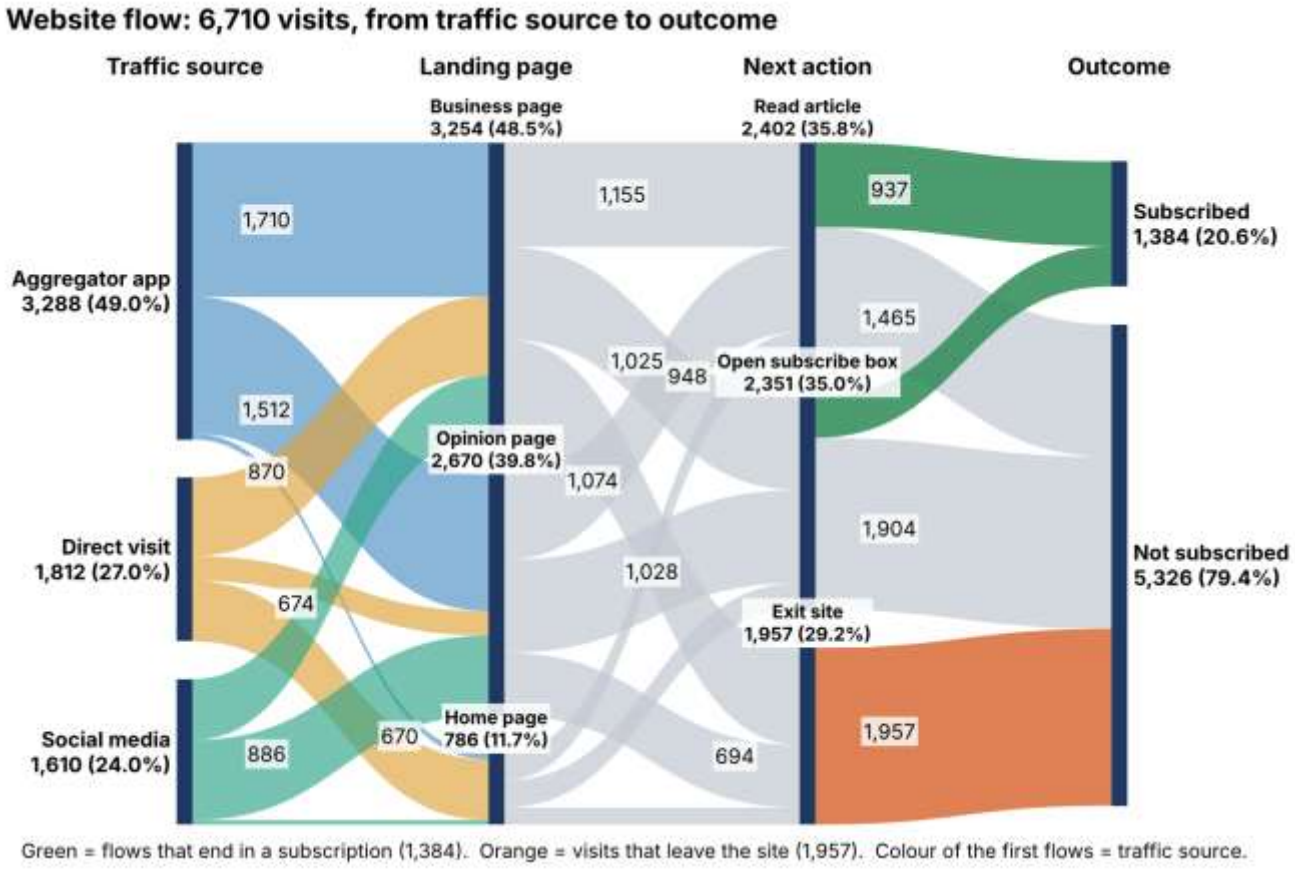
Step 3: Check: the other two flows from Opinion page are 1,028 and 694, and 948 + 1,028 + 694 = 2,670.

Result: About 35.5% of the visits that land on the Opinion page go on to read an article. The other 64.5% open the subscribe box (38.5%) or leave the site (26.0%).

(c) Design of the Final Sankey [4 marks]

Aim: To give four design decisions for the final diagram, each with a one-line reason taken from the data.

Final Diagram



Four Design Decisions

Decision	What I did	Why (from this data)
1. Node order	In every column the nodes are placed largest first (Aggregator app, Direct visit, Social media; Business, Opinion, Home; Read article, Open subscribe box, Exit site; Subscribed above Not subscribed). The bands leaving a node are stacked in the same order as their targets.	I counted the crossings for all 432 possible node orders (6 x 6 x 6 x 2), and this order has the fewest crossing pairs of flows (19). Stacking the bands by target stops them from twisting over their neighbours.
2. Colour rule	The first flows are coloured by traffic source (blue, yellow, teal). Flows between the middle stages are light grey. Only two other colours are used: green for flows that end in Subscribed and orange for the flow that leaves the site.	Traffic source is the one thing that explains where visits come from, and 3 colours are easy to tell apart. Grey keeps the 9 middle flows quiet, so the green and orange flows stand out.
3. Labels	Every node shows its name, its visits and its share of the 6,710 visits. Flows of 500 visits or more carry their number. The three hidden flows now carry their calculated values, and no "?" is left.	Percentages such as 20.6% Subscribed and 79.4% Not subscribed answer the main question at a glance. Skipping the numbers on the smallest flows (66, 50, 189, 298, 299, 447) stops labels from overlapping.
4. Highlight	The orange Exit site to Not subscribed band (1,957) and the green bands into Subscribed (937 + 447 = 1,384) are the only strong colours, and the title states the message.	The biggest single loss is 1,957 visits (29.2%) who leave the site, while only 1,384 (20.6%) subscribe. These two flows are the story the analytics team has to act on.

Result: The final Sankey keeps the largest-first order with the fewest crossings, uses colour only for the traffic source and for the two outcomes that matter, labels every node with its share, and highlights the 1,957 visits that leave the site.

(d) Printed Node Heights [2 marks]

Aim: To find the height in cm of the three landing-page nodes when 1 cm of node height stands for 300 visits.

Working

Step 1: Height in cm = visits / 300, because 300 visits are drawn as 1 cm.

Step 2: Opinion page: 2,670 / 300 = 8.90 cm.

Step 3: Business page: 3,254 / 300 = 10.8467, which is 10.85 cm to two decimal places.

Step 4: Home page: 786 / 300 = 2.62 cm.

Step 5: Check: the three heights add up to 8.90 + 10.85 + 2.62 = 22.37 cm, which is the same as 6,710 / 300 = 22.37 cm for all visits in the column.

Result

Landing-page node	Visits	Calculation	Height (cm)
Opinion page	2,670	2,670 / 300	8.90
Business page	3,254	3,254 / 300	10.85
Home page	786	786 / 300	2.62

The nodes of one column need 22.37 cm in total, plus the gaps between them, so the diagram fits on an A4 sheet (29.7 cm high) if the gaps are kept small.

(e) R Code with networkD3 [4 marks]

Aim: To write a links data frame, a nodes data frame and the sankeyNetwork() call that draws the whole diagram.

R Program

```
# Links: first-stage flows, then landing page -> next action, then next action -> outcome
nodes <- data.frame(name = c(
  "Aggregator app", "Direct visit", "Social media",       # 0-2  traffic source
  "Business page", "Opinion page", "Home page",           # 3-5  landing page
  "Read article", "Open subscribe box", "Exit site",      # 6-8  next action
  "Subscribed", "Not subscribed"))                        # 9-10 outcome

links <- data.frame(
  source = c(0,0,0, 1,1,1, 2,2,2,   3,3,3, 4,4,4, 5,5,5,   6,6, 7,7, 8),
  target = c(3,4,5, 3,4,5, 3,4,5,   6,7,8, 6,7,8, 6,7,8,   9,10, 9,10, 10),
  value  = c(1710,1512,66, 870,272,670, 674,886,50,
             1155,1025,1074, 948,1028,694, 299,298,189,
             937,1465, 447,1904, 1957))

# The diagram
library(networkD3)
sankeyNetwork(Links = links, Nodes = nodes,
              Source = "source", Target = "target", Value = "value",
              NodeID = "name", units = "visits",
              fontSize = 14, nodeWidth = 25, sinksRight = TRUE)
```

Steps

Step 1: The nodes data frame lists the 11 node names once, in the order of the diagram (traffic source, landing page, next action, outcome). The row position of a name is its node number, and networkD3 counts from 0, so Aggregator app is 0 and Not subscribed is 10.

Step 2: The links data frame has one row for each flow with three columns: source (where the flow starts), target (where it ends) and value (the visits). The 23 rows follow the table order of the diagram:

first the nine flows from the traffic sources (rows 4 to 9 are the six leaving Direct visit and Social media), then landing page to next action, then next action to outcome.

Step 3: The three hidden flows are filled with the values calculated in part (a): 870 (Direct visit to Business page), 189 (Home page to Exit site) and 1,465 (Read article to Not subscribed).

Step 4: sankeyNetwork() takes the two data frames and the names of their columns: Source, Target and Value from links and NodeID from nodes. units = "visits" puts the word "visits" in the tooltip, nodeWidth sets the width of the node bars and sinksRight = TRUE keeps the outcome nodes on the right edge.

Result: The two data frames and the single sankeyNetwork() call draw the whole four-stage diagram of 6,710 visits. The sum checks in R agree with every node total in the question, which confirms the three hidden values.

(f) When a Sankey Misleads [2 marks]

Aim: To state two situations in which a Sankey diagram misleads or becomes unreadable, and to name an alternative chart for each.

Answer

Situation	Why the Sankey fails	Alternative chart
1. Too many nodes and flows (for example every page of the website, with hundreds of small flows)	The bands cross each other many times, the small ones become hairlines that cannot be labelled, and the picture turns into tangled spaghetti. Even here, with only 11 nodes, the middle stage has 9 crossing flows.	A heatmap (matrix) of source against destination, where the colour shows the visits, or a sorted bar chart of only the top 10 flows.
2. The reader has to compare exact sizes of flows (for example 1,028 against 1,074 visits)	Bands are slanted, curved and overlap, so their widths are hard to compare, and the eye cannot tell small differences. Sankey is also misleading when the node heights are not drawn on one common scale, or when the flows are not conserved.	A sorted horizontal bar chart (or a stacked bar for each stage), where the length from a common baseline is compared.

Result: A Sankey diagram suits a small number of nodes with the question "where does the total go?". It misleads when there are too many flows (use a heatmap or a bar chart of the top flows) and when exact comparison is needed (use a sorted bar chart).

Summary of Answers

Part	Answer

(a)	Read article to Not subscribed = 1,465; Home page to Exit site = 189; Direct visit to Business page = 870
(b)	(i) 20.63%; (ii) 1,957 (Exit site to Not subscribed, 29.2% of visits); (iii) 35.51%
(c)	Largest-first order with the fewest crossings, colour by traffic source with green and orange highlights, labels with shares, highlight on the 1,957 leaving visits
(d)	Opinion page 8.90 cm; Business page 10.85 cm; Home page 2.62 cm
(e)	links and nodes data frames with sankeyNetwork() (networkD3)
(f)	Too many flows: use a heatmap or top-flow bar chart; exact comparison: use a sorted bar chart

Conclusion

The rule that what enters a node equals what leaves it gave all three hidden flows and confirmed every node total. Of the 6,710 visits, only 20.6% end in a subscription, and the biggest single loss is the 1,957 visits that leave the site without subscribing. A Sankey diagram is the right chart for showing this flow when it has a few nodes, but it must be kept simple, ordered to cut crossings and highlighted at the points that matter.